

## Executive Summary

【18†embed\_image】 *Figure: Example INCOIS PFZ advisory map (red/blue lines show Potential Fishing Zones for different sectors). INCOIS publishes daily PFZ maps and text per coastal sector* 【3†L115-L123】 【3†L129-L133】 . Potential Fishing Zone (PFZ) advisories are generated **daily for 14 coastal sectors** of India using satellite SST and chlorophyll data 【3†L112-L119】 . Each advisory includes a map and a text list of PFZ locations (latitude, longitude), depth, distance and direction from nearby landing centers 【3†L129-L133】 . The **INCOIS PFZ web portal** (TextDataHome) provides the current advisories by sector, but no simple “download all” link is given. However, INCOIS’s *Data Holdings* site notes that PFZ advisories from 2003–present are available in ASCII/JPEG format 【13†L142-L144】 (likely via data request). In practice, one can programmatically scrape or query the INCOIS PFZ interface for today’s advisories, and/or request bulk archive data. To build a machine-learning dataset, we extract PFZ point coordinates and labels from these advisories, then join them with environmental covariates (SST, chlorophyll, wind, bathymetry) from NOAA/NASA/GEBCO. The workflow is:

```
flowchart TB
    A[INCOIS PFZ Text/ASCII] --> B[Parse PFZ coordinates (lat, lon, depth, etc.)]
    C[Satellite Data (SST, Chl, Wind, Bathy)] --> D[Extract environmental features]
    B --> E[Merge PFZ labels + features]
    D --> E
    E --> F[ML-ready dataset (CSV/Parquet)]
```

## PFZ Advisories – Sources & Formats

- **INCOIS PFZ Web Interface (Text):** Current PFZ forecasts are available interactively by sector (14 sectors) on the INCOIS PFZ web page 【3†L115-L123】 . This delivers a table of PFZ points (lat, lon, depth, distance, direction). There is no public bulk URL, but one can simulate form selections for each sector/language. (Each advisory is for “Forecast Date = today” and “Valid up to = tomorrow” on the site, e.g. 29 Aug 2026 → 30 Aug 2026 【17†L77-L80】 .)
- **INCOIS PFZ WebGIS:** An interactive Leaflet-based map (WebGIS) shows *all* PFZ points along India’s coast ( $\approx 1223$  nodes) 【37†L70-L78】 . Layers include SST, chlorophyll, EEZ, sectors, landing centers, bathymetry, and the PFZ advisory line. This is a web client (WMS/WMTS tiles) – no direct REST API, but coordinates could be retrieved by inspecting GIS layers if accessible.
- **INCOIS Data Holdings:** The official catalog lists **PFZ Advisories** (ASCII/JPEG) from 2003 to date 【13†L142-L144】 . This suggests that historical PFZ text (ASCII) and map images (JPEG) are archived. Bulk download is not directly open; one must submit a data requisition form on the INCOIS portal 【13†L142-L144】 . If granted, the data likely come as ASCII tables or zipped files (naming conventions not public).

- **ESSDP/MOES Portal:** No specific PFZ dataset is exposed via ESSDP. (ESSDP mostly covers gridded reanalysis and research data, not daily advisories.)
- **Third-Party Sources:** None known. (PFZ is an India-specific product. One could use satellite data like GHRSSST for SST or MODIS for chlorophyll independently.)

| Source/Portal                  | Content                                  | Format                                         | Range                  | Access                 | Notes                                                                                             |
|--------------------------------|------------------------------------------|------------------------------------------------|------------------------|------------------------|---------------------------------------------------------------------------------------------------|
| INCOIS PFZ Web Interface       | Today's PFZ points by sector             | HTML/Text (interactive)                        | Daily (near-real-time) | Public (web)           | No bulk link; can scrape or simulate form (by sector/lang).                                       |
| INCOIS PFZ WebGIS              | PFZ geo-locations & layers (current)     | WMS/WMTS (tiles), likely GeoJSON/WFS available | Daily (current)        | Public (web)           | Interactive map layers for PFZ, SST, chl, etc.                                                    |
| INCOIS Data Holdings (PFZ)     | Historical PFZ advisories                | ASCII / JPEG                                   | 2003 – present         | Registration (request) | Requires data requisition; format = ASCII text or zipped tables <a href="#">【13†L142-L144】</a> .  |
| NOAA GHRSSST (SST)             | High-res SST (used as input)             | NetCDF (OPeNDAP/FTP)                           | 1982 – present         | Public (noaa.gov)      | E.g. NOAA OISST V2, 0.25° daily SST <a href="#">【13†L158-L164】</a> .                              |
| NASA MODIS/ Aqua (Chlorophyll) | Ocean color Chlorophyll                  | NetCDF/HDF (OPeNDAP)                           | 2002 – present         | Public (oceancolor)    | e.g. MODIS-Aqua monthly/daily chlorophyll (MODISA).                                               |
| Copernicus ERA5 (Wind)         | Surface wind, other reanalysis variables | NetCDF (Copernicus)                            | 1979 – present         | Public (ECMWF)         | ERA5 hourly/daily winds (supported by Python <code>cdsapi</code> ).                               |
| GEBCO 2026 (Bathymetry)        | Global ocean depth/ bathymetry           | NetCDF (15")                                   | static global          | Public (GeBCO)         | Global bathymetry grid, public domain <a href="#">【95†L9-L13】</a> <a href="#">【95†L98-L101】</a> . |

## PFZ Text/ASCII Download (Daily)

To retrieve today's PFZ data in text form, one approach is to simulate the sector selection on INCOIS's **Marine Fisheries TextDataHome** interface. For example, using Python's `requests` (or `selenium`) to submit the form for a given sector and language:

```
import requests
from bs4 import BeautifulSoup

session = requests.Session()
# Example: get PFZ text for Sector=4 (Karnataka) in English
params = {'mfid': 1, 'sector': 4, 'lang': 'English', 'request_locale': 'en'}
url = "https://incois.gov.in/MarineFisheries/TextDataFetch" # hypothetical endpoint
res = session.get(url, params=params, verify=False)
soup = BeautifulSoup(res.text, 'html.parser')

# Extract table or text from HTML (structure may vary)
pfz_div = soup.find("div", {"id": "pfzText"})
pfz_text = pfz_div.get_text(separator="\n")
print(pfz_text)
```

In practice, the exact endpoint and form fields may require experimenting or capturing network calls. Once the HTML is fetched, the text contains PFZ entries like:

```
12.45 74.82 45m 42km NW Honnavar
12.50 74.90 50m 38km NW Bhatkal
...
```

Each line has *latitude, longitude, depth, distance, direction, landing centre*. A `pandas` DataFrame can be built from these columns for further processing.

One can also use `curl` in bash to fetch and pipe into Python. For example, with an authenticated session cookie (if needed) one might do:

```
curl -k "https://incois.gov.in/MarineFisheries/TextDataFetch?
sector=4&lang=English" \
-o pfz_karnataka.txt
```

(Replace `sector` index and `lang` as needed.) Alternatively, use Selenium to control a headless browser:

```

from selenium import webdriver
driver = webdriver.Chrome()
driver.get("https://incois.gov.in/MarineFisheries/TextDataHome?mfid=1")
# Select sector/language in the form
# ... (Selenium code to select option and submit)
pfz_text = driver.find_element_by_id("pfzText").text

```

After fetching the text, parse lines and extract fields into a CSV or DataFrame (see next section).

## PFZ Bulk Archive (Historical)

For **historical PFZ advisories (2003–present)**, INCOIS does not provide a direct download link. The Data Holdings page indicates the data exist (ASCII/JPEG) but access is by request 【13†L142-L144】. In practice, one must fill out the INCOIS *Data Requisition Form* (accessible from [Data Holdings] or [Data Requisition] menu) specifying “PFZ Advisories” for the desired date range. The system will then deliver a batch of ASCII files (or zipped archives) for the requested period.

If a direct URL were known, it might look like:

```
https://incois.gov.in/data/MarineFisheries/PFZ_ADVISORY/YYYYMMDD_sector4.txt
```

but no public pattern is documented. Therefore, automated bulk downloads generally involve contacting INCOIS. Once obtained, each ASCII file (per sector per day) can be opened as plain text for parsing.

## Parsing PFZ Advisories to CSV

Whether obtained via scraping or manual download, the PFZ advisory text has a consistent format. For example, a text snippet for Sector 4 (Karnataka) might be:

```

PFZ for KARAIKAL (Sector 4) valid up to 30 Aug 2026
Latitude Longitude Depth Distance Direction Landing Centre
12.45 74.82 45m 42km NW Honnavar
12.50 74.90 50m 38km NW Bhatkal
...

```

We extract rows where latitude/longitude appear. A Python code snippet to parse this into structured data:

```

lines = pfz_text.splitlines()
records = []
for line in lines:
    parts = line.split()
    if len(parts) >= 5 and parts[0].replace('.', '', 1).isdigit():

```

```

lat = float(parts[0]); lon = float(parts[1])
depth = parts[2]; dist = parts[3]; direction = parts[4]
landing = " ".join(parts[5:])
records.append((lat, lon, depth, dist, direction, landing))

import pandas as pd
df = pd.DataFrame(records, columns=['Lat', 'Lon', 'Depth', 'Dist', 'Dir', 'Landing'])
df['Date'] = '2026-08-29' # example date of forecast
df.to_csv('pfz_sector4_20260829.csv', index=False)

```

This produces a CSV with columns (for ML):

Date, Sector, Lat, Lon, Depth, Dist\_km, Direction, Landing. We would repeat for all sectors/dates and concatenate into one master table.

## Extracting from PFZ WebGIS or GeoJSON

INCOIS's **PFZ WebGIS** layers (via `DataInfo/MFASPFZ/index.html` or `geoportal/MFASPFZ/index.html`) include PFZ as GIS features. If one can discover the underlying GeoServer/WFS endpoints, one could use `requests` or `geopandas` to fetch them. For example, if a WFS service were available:

```

import requests
from io import BytesIO
from zipfile import ZipFile

wfs_url = "https://incois.gov.in/geoserver/wfs?
service=WFS&version=1.0.0&request=GetFeature&typeName=PFZ_Shapes&outputFormat=shape-
zip"
res = requests.get(wfs_url, verify=False)
z = ZipFile(BytesIO(res.content))
# Use geopandas to read the shapefile inside
import geopandas as gpd
gdf = gpd.read_file(z.open('PFZ_Shapes.shp'))

```

(The above is illustrative; the actual `typeName` and URL would need to be discovered via INCOIS's GIS configuration.) The result would be a GeoDataFrame with each PFZ polygon or point and its attributes, which we can convert to a CSV of coordinates and metadata.

## Supporting Environmental Data

To build a feature-rich ML dataset, each PFZ point is augmented with environmental covariates at that location and time:

- **Sea Surface Temperature (SST)** – e.g. NOAA OISST (0.25° global daily SST). NOAA provides these via OPeNDAP. INCOIS itself archives AVHRR SST (1986–) `【11†L124-L130】` `【13†L158-L164】`. Example code using Python's `xarray`:

```
import xarray as xr
# NOAA OISST via OPeNDAP
ds = xr.open_dataset("https://www.ncei.noaa.gov/thredds/dodsC/OisstBase/
NetCDF/AVHRR/2026/AVHRR_20260829.nc")
sst_val = ds['sst'].sel(lat=12.45, lon=74.82, method="nearest").item()
```

(Replace path with the proper NOAA URL; each dataset covers one day.)

- **Chlorophyll** – e.g. NASA MODIS-Aqua daily/8-day chlorophyll (global 4km). Accessible via OceanColor or Copernicus services. In Python, one can use OPeNDAP or the `netCDF4` library to read files. Alternatively, use Copernicus Marine: e.g. C3S satellite chlorophyll reanalysis (SST/Chl blending).
- **Surface Wind** – e.g. ERA5 reanalysis from ECMWF (hourly, 0.25°). Using `cdsapi`:

```
import cdsapi
c = cdsapi.Client()
c.retrieve("reanalysis-era5-single-levels", {
    "product_type": "reanalysis",
    "variable": "10m_u_component_of_wind",
    "date": "2026-08-29",
    "area": [30, 65, 8, 95], # N, W, S, E (approx India)
    "format": "netcdf"
}, "wind_u_20260829.nc")
```

Or simpler: use global GFS or NCEP reanalysis via NOAA.

- **Bathymetry/Depth** – e.g. GEBCO (latest 15" global grid [95†L9-L13] ). Python code to sample:

```
ds_bathy = xr.open_dataset("https://www.gebco.net/data/gebco_2026.nc")
depth = ds_bathy['elevation'].sel(lat=12.45, lon=74.82,
method="nearest").item()
```

GEBCO is in the public domain [95†L98-L101] .

These sources cover the same period as PFZ advisories. We join the environmental values by matching date and nearest grid point to each PFZ coordinate.

## Code Example: Merging into CSV

A simplified workflow (using pandas/xarray) to combine PFZ labels with SST and chlorophyll:

```
import pandas as pd, xarray as xr

# Load PFZ points (example)
```

```

pfz_df = pd.read_csv('pfz_allsectors_20260829.csv') # has Date,Lat,Lon,...

# Load environmental data for that date
sst_ds = xr.open_dataset('https://.../AVHRR_20260829.nc') # NOAA OISST
chl_ds = xr.open_dataset('https://.../MODISA_20260829.nc') # hypothetical URL
wind_ds = xr.open_dataset('https://.../ERA5_20260829.nc') # U-component etc.
bathy_ds = xr.open_dataset('https://www.gebco.net/data/gebco_2026.nc')

# Function to get nearest values
def get_val(ds, var, lat, lon):
    return ds[var].sel(lat=lat, lon=lon, method="nearest").item()

features = []
for idx, row in pfz_df.iterrows():
    lat, lon = row['Lat'], row['Lon']
    sst = get_val(sst_ds, 'sst', lat, lon)
    chl = get_val(chl_ds, 'chlor_a', lat, lon)
    wind = get_val(wind_ds, 'u10', lat, lon) # e.g.
    depth = get_val(bathy_ds, 'elevation', lat, lon)
    features.append((sst, chl, wind, depth))
pfz_df[['SST', 'Chl', 'Wind', 'Depth_model']] = features
pfz_df.to_csv('pfz_dataset_20260829.csv', index=False)

```

This produces rows like (Date, Sector, Lat, Lon, Depth\_obs, Dist\_km, Dir, Landing, SST, Chl, Wind, Depth\_model).

## Data Schema and Sample

A typical ML-ready schema (CSV) might be:

| Date       | Sector | Lat   | Lon   | Depth_obs | Dist_km | Dir | Landing  | SST    | Chl  | Wind | Bathy |
|------------|--------|-------|-------|-----------|---------|-----|----------|--------|------|------|-------|
| 2026-08-29 | 4      | 12.45 | 74.82 | 45        | 42      | NW  | Honnavar | 28.7°C | 0.35 | 5.2  | -112  |
| 2026-08-29 | 4      | 12.50 | 74.90 | 50        | 38      | NW  | Bhatkal  | 28.9   | 0.30 | 5.1  | -114  |
| 2026-08-29 | 4      | 12.55 | 74.95 | 60        | 35      | NW  | Bhatkal  | 29.1   | 0.28 | 4.9  | -118  |
| 2026-08-29 | 4      | 12.60 | 75.00 | 55        | 30      | NW  | Ankola   | 28.8   | 0.31 | 5.0  | -120  |
| 2026-08-29 | 4      | 12.65 | 75.05 | 50        | 25      | NW  | Karwar   | 28.6   | 0.33 | 5.3  | -110  |

(Example data for Karnataka on 29 Aug 2026. SST in °C, Chl in mg/m<sup>3</sup>, Wind in m/s, Bathy from GEBCO).

## Licensing and Access

INCOIS data are government-compiled. **PFZ advisories** appear non-sensitive and intended for public benefit, but archival data typically require an official request. The Data Holdings page lists PFZ as “Public

Access with download options” [13†L142-L144] , implying no strict paywall, but formal procedure. There is no explicit license, so cite INCOIS when using.

**GEBCO bathymetry** is public domain [95†L98-L101] (free for any use). **NOAA** and **NASA** data are open (usually “free use, cite NOAA/NASA”). **ERA5** (via Copernicus) is free for research with proper attribution. The merged dataset can be shared following these policies.

## Download Options Summary

| Portal                        | Data                               | Format             | Dates              | Access                    | Notes                                                        |
|-------------------------------|------------------------------------|--------------------|--------------------|---------------------------|--------------------------------------------------------------|
| INCOIS PFZ<br>TextData page   | PFZ advisory<br>(current)          | HTML/<br>ASCII     | Daily<br>(today)   | Public (web)              | Need to script form;<br>one sector at a<br>time.             |
| INCOIS PFZ<br>WebGIS          | PFZ geo-<br>locations<br>(current) | WMS/<br>GeoJSON?   | Daily<br>(current) | Public (web)              | Interactive map;<br>could extract via<br>WFS/GeoJSON.        |
| INCOIS Data<br>Holdings (PFZ) | PFZ advisories<br>(archive)        | ASCII, JPEG        | 2003–<br>present   | Request<br>(registration) | Bulk historical data<br>via request form<br>[13†L142-L144] . |
| NOAA OISST<br>(e.g. GHR SST)  | SST (ocean)                        | NetCDF/<br>OPeNDAP | 1982–<br>present   | Public<br>(noaa.gov)      | Gridded daily SST<br>(0.25°).                                |
| NASA<br>OceanColor<br>(MODIS) | Chlorophyll<br>(ocean)             | NetCDF/<br>HDF     | 2002–<br>present   | Public<br>(oceancolor)    | 4-km daily/8-day<br>chlorophyll.                             |
| Copernicus<br>Marine (ERA5)   | Wind, etc.                         | NetCDF<br>(CDS)    | 1979–<br>present   | Public<br>(ECMWF)         | Hourly/daily<br>reanalysis via CDS<br>API.                   |
| GEBCO (15”<br>bathymetry)     | Global depth<br>model              | NetCDF             | static             | Public<br>(gebco.net)     | Global grid with<br>undersea depth<br>[95†L9-L13] .          |

Each source above provides APIs or FTP/OPeNDAP for automation. For example, NOAA’s OISST can be accessed via NCEI THREDDS, and GEBCO offers netCDF downloads [95†L9-L13] .

In summary, **PFZ labels** come from INCOIS (web/API or archives) [3†L129-L133] [13†L142-L144] , and **features** come from standard open ocean datasets. Together they form an ML-ready dataset to predict PFZ occurrence along the Indian coast.

**Sources:** INCOIS PFZ documentation [3†L115-L123] [3†L129-L133] [13†L142-L144] , INCOIS Data Holdings [13†L142-L144] [11†L124-L130] , GEBCO description [95†L9-L13] [95†L98-L101] , NOAA/ NASA dataset docs (as noted).